

Christi Sanderson

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PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer (Java & AI Integration) with 14+ years of experience at FedEx specializing in scalable, event-driven **Java Spring Boot microservices** architecture, high-throughput distributed systems, batch migrations, and legacy modernizations. Proven track record of delivering business-critical software for fast-paced **startups** and **enterprise** environments such as FedEx, Disney and Lockheed Martin. Independently expanded skills to full-stack design and development of advanced **AI/ML** technologies including observable multi-agent workflows (**Langfuse, LangGraph**), asynchronous orchestration (**PyTorch**), classic and deep machine learning (**Scikit-learn, TensorFlow**), **open-source LLMs, declarative UIs, RAG** pipelines, and scalable **Python** backends. Designed and implemented a hybrid, event-driven **AI document extraction** and **PII redaction** engine using React, **Node.js, Spring Boot**, and **Python AI services** integrated with **AWS serverless** infrastructure (**S3, SQS, Lambda**), PostgreSQL/**pgvector**, **OCR, NLP, RAG, Ollama LLMs, Resilience4j** for scalable and fault-tolerant document processing, and containerization via Docker Compose.

TECHNICAL SKILLS

- **AI & Orchestration:** LangGraph, LangChain, Multi-Agent Workflows, RAG, Ollama, Langfuse, Prompt Engineering, Copilot, Claude, VS Code
- **Document AI:** PyMuPDF, Tesseract OCR, spaCy NER redaction, pgvector, NLP, Structured LLM Extraction
- **Core Machine Learning:** PyTorch, TensorFlow, Scikit-learn, MobileNetV2, Text-to-SQL
- **Data Architecture:** Oracle, Datastax Cassandra, Qdrant, ChromaDB, PostgreSQL, Pydantic, JSON, Schema Validation
- **Backend Systems:** Java Spring Boot Microservices, Python, FastAPI, Node.js, REST APIs, Resilience4j, JMS Asynchronous Messaging
- **Frontend & UI:** React, Angular, JavaScript/TypeScript, Tailwind CSS, Declarative UI, JSON Schema
- **DevOps:** Docker, Docker Compose, CI/CD Pipelines, GitHub, AppD, Splunk, Dynatrace
- **Cloud:** AWS (Lambda, SQS, S3), LocalStack, GCP, Azure, Cloud Foundry

PROFESSIONAL EXPERIENCE

FedEx Services | Orlando, FL

Full Stack Developer III | January 2012 – June 2026

- **Awarded** prestigious **FedEx Bravo Zulu Awards** for above and beyond performance resulting in user interface performance improvements, revenue cost reductions, operational efficiencies, and enterprise PCI compliance and authentication solutions.
- **Designed** and built scalable, fault-tolerant **Java Spring Boot microservices** featuring service discovery and API gateway to execute concurrent batch processing across multiple global document repositories.
- **Enhanced** Enterprise Document Management core web services to enable client adoption resulting in **7% revenue growth** targeting an **\$8M revenue increase**.

- **Agile Team Lead** that **saved \$1.2M annually in licensing costs and increased operational efficiencies** by migrating vendor document repositories to the core FedEx platform.
- **Integrated AI tooling into production support**, accelerating systemic defect resolution and automating testing frameworks.
- **Developed Google Cloud Storage (GCS)** bulk export scripts to support compliance, audit research, and legal document reporting.
- **Engineered** React and Node.js data engines to visualize real-time system metrics, providing leadership with actionable performance dashboards.
- **Engineered** a high-throughput **Java** migration utility to safely transition 500M+ legacy documents into **Oracle, Datastax, and Google Cloud Storage (GCS)**.
- **Validated Datastax** conversion models by developing a Java Spring Boot **proof-of-concept (POC)** using embedded **Cassandra** databases with **Spring JPA**.
- **Achieved full PCI compliance** for high-volume global airway bill processing by partnering with InfoSec to implement enterprise eKMS encryption APIs and implementing custom SSL layers.
- **Modernized** legacy applications to **REST services** architecture while introducing **CI/CD automation** pipelines and metrics dashboards.

PERSONAL AI/ML ENGINEERING PROJECTS

Multi-Agent Research Approval Assistant

- **Lowered operational bottlenecks by 35%** using LangGraph stateful workflows with human-in-the-loop approvals and Langfuse observability in Docker for trace monitoring.
- Designed resilient, stateful, and resumable execution loops for advanced agent management.

Enterprise Autonomous Analytics & Knowledge Engine

- **Achieved zero API costs, complete data privacy, and reduced hallucinations** by deploying a local Llama 3 engine with deterministic guardrails.
- Engineered a full-stack platform using decoupled microservices, Qdrant RAG pipelines, and PostgreSQL Text-to-SQL with React and Node.js frontends for deterministic UI layout rendering.

OmniMind: Multi-Framework AI Agent Ecosystem

- **Improved multi-modal task routing speed by 50%** by building an asynchronous orchestration engine that unifies Scikit-learn, TensorFlow, and LangChain plus Ollama.
- Integrated tabular analytics and MobileNetV2 vision inference into a single scalable agent workflow.

RAG Research Assistant

- **Reduced document retrieval time and user-perceived latency by 40%** by deploying a local LLM vector retrieval system and streaming token responses via Ollama and LangChain.
- Developed a full-stack general and semantic search chatbots backed by FastAPI.

Asynchronous Secure Document Intelligence Pipeline | *Hybrid Event-Driven AI Architecture*

- **Architected a hybrid event-driven platform** by combining LocalStack-emulated AWS services with containerized Node.js, Spring Boot, and Python AI microservices.
- **Cut cloud development costs by 100%** during testing by building a local AWS simulation using LocalStack S3, SQS, and Lambda, saving an estimated **\$1,500/month** in infrastructure fees.
- **Implemented secure data ingestion** via direct PDF uploads to S3, S3 ObjectCreated event triggers, Lambda metadata extraction, and asynchronous SQS task publishing.
- **Decoupled user-facing ingestion** from intensive AI analysis using SQS queues, reducing API response times by **85%** and utilizing visibility timeouts and dead-letter queues (DLQ) for **zero-loss error handling**.
- **Developed a Python AI service** using PyMuPDF, Tesseract OCR, spaCy NER redaction, LangChain chunking, pgvector, and Ollama, achieving privacy-aware document summarization in **under 4 seconds** per page.
- **Integrated a Java Spring Boot worker** for SQS consumption, compliance validation, S3 retrieval, AI orchestration, and PostgreSQL persistence, leveraging Resilience4j circuit breaking to maintain **99.9% uptime** during simulated microservice failures.
- **Containerized the platform with Docker Compose** and designed production-ready AWS deployment configurations using Terraform, ECS/Fargate, and Aurora PostgreSQL, engineered to scale automatically to handle over **10,000 document uploads per day**.

PREVIOUS PROFESSIONAL EXPERIENCE

Walt Disney World | Lake Buena Vista, FL

Technical Specialist

- **Developed** enterprise **Java** enhancements and data **workflows** for publish to core sales and booking composite system channels.
- **Engineered predictive recommendation** modules to process and store high-volume scoring data generated by Revenue Management engines to drive the strategic products and services offers presented for sale on the public internet.

M2 Systems Corporation (Muscato Corporation) | Orlando, FL

Senior Software Engineer

- **Developed** multi-platform online transaction processing (OLTP) switch interfaces connecting disparate systems in healthcare and debit networks. Most notable clients included Blue Cross/Blue Shield and Sacred Heart Medical Center.
- **Collaborated with CEO** and product team during development at customer site.

TriCom, Inc. (ISR Global Telecom) | Orlando, FL

Embedded Systems Engineer

- **Served** as a **founding** Embedded Systems Engineer in a 6-employee **spin-off** to **develop** line protocol task conversion software.
- **Deployed** independently to **Germany** to rollout production software and **execute** rigorous Acceptance Testing at the DaimlerChrysler corporate facility.

Lockheed Martin Corporation | Orlando, FL

Systems Engineer

- **Selected for Engineering CoOp program** leveraging advanced mathematical modeling.
- **Developed** simulations to analyze navigation, tracking, and fixed-point scaling algorithms.
- **Commended** for **designing** a ground-based FLIR/LTT turret simulation to evaluate air-target feasibility.
- **Digitized** missile subsystem block diagrams (airframe, actuator, sensors, feedback loops) and tuned IMU filters via fly-out sensitivity and transfer alignment analyses.

CERTIFICATIONS & TRAINING

Udacity AI Tech Practitioner Training - Generative AI Fundamentals

Achieved Accomplished AI Solutions Architect Rating

GitHub Copilot Training & AI Engineering Workshops

FedEx Corporate Training

Mobile Application Development Certification

University of California, Irvine

EDUCATION

Bachelor of Science, Computer Science (Scientific Specialization), Cum Laude

Minor: Business Administration

University of Central Florida, Orlando, FL

Academic Scholarship, Time-Shortened Degree